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Gender Equality and Diversity in Engineering Ethics:
A Case Study of Hiring and Promotion Bias in a Safety-Critical Project

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Date:2025/11/25

Title:

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Introduction:

This case study examines gender equality and diversity as an ethical issue within engineering practice. Gender imbalance, biased hiring and promotion, and exclusionary workplace cultures are not only social justice problems but also engineering problems: they affect team performance, innovation, and ultimately public safety and welfare. The purpose of this case study is to analyze an ethical dilemma arising from systemic gender bias in hiring/promotion at an engineering firm working on a safety-critical product, evaluate stakeholders' responsibilities, apply ethical theories and professional codes, and produce recommendations for ethically defensible action. [SDGs+1](#)

Background / Context :

Scenario (hypothetical but realistic). Mid-sized engineering firm "NovaTech" is developing an embedded control system for a new consumer medical device (insulin pump). Project timelines are tight and regulatory approval depends on robust testing and documentation. NovaTech's engineering division has historically been male-dominated at senior levels: of 60 engineers, 48 identify as male and 12 as female; of 10 senior leads, only 1 is female.

Events that trigger the dilemma. During recruitment and promotion cycles, qualified female applicants and mid-level female engineers report being passed over for promotions and high-visibility assignments (e.g., control-algorithm lead) in favor of less-qualified or less-experienced male colleagues. Exit interviews reveal that women feel excluded from technical discussions and mentoring programs. A recent product test failure was later traced partly to incomplete test scenarios that would likely have been caught by a more diverse design review team.

Project / system details. The insulin pump's firmware interacts with hardware safety interlocks and must meet strict IEC/ISO standards for medical devices; human factors and inclusive testing are critical. Missing perspectives in requirements and testing can lead to unanticipated failure modes that affect patient safety.

Key stakeholders.

- ❖ Affected engineers (female and male staff), applicants, and potential applicants.
- ❖ NovaTech management and HR.
- ❖ The company's clients (medical device manufacturer).
- ❖ End users (patients relying on the device), caregivers, and healthcare providers.
- ❖ Regulators and standards bodies.
- ❖ Society at large (public trust in engineering products). asme.org+1

Ethical Dilemma:

Conflict / decision point. NovaTech's leadership must decide how to respond to evidence of gender-biased hiring/promotion practices that appear to be systemic. The decision point: Should management (A) treat the problem as primarily a human-resources/talent pipeline issue and make incremental, voluntary changes (mentoring, unconscious bias training), (B) implement immediate structural changes (anonymized hiring, affirmative targets for leadership, transparent promotion criteria, reassignments), or (C) do little or nothing and prioritize schedule and budgets over corrective measures?

Possible options and potential consequences.

- ❖ Option A — Incremental, voluntary measures (training, mentoring).
Pros: Low disruption; signals intent to change.

Cons: May be insufficient to correct systemic bias; slow results; continued exclusion risks further product-quality issues, legal exposure, and reputational harm. Evidence shows that surface measures without accountability often fail to produce equitable outcomes. [McKinsey & Company](#)

- ❖ Option B — Structural interventions (blind screening, transparent metrics, targets, reorganize assignments).

Pros: Faster, measurable change; reduces disparate impact; enhances team diversity, innovation, and detection of design faults; aligns with professional codes and SDG commitments.

Cons: Short-term disruption, possible internal resistance or legal/PR misinterpretation if perceived as “quotas”; requires investment to implement fairly.

- ❖ Option C — Do nothing / prioritize schedule and cost.

Pros: Short-term schedule preservation.

Cons: Ethical failure to protect public welfare and employee rights; increased risk of product faults, loss of talent (especially women), legal liability, and harm to users and public trust—contradictory to engineering professional duties. [nspe.org+1](#)

Moral Sensitivity:

How the ethical problem was recognized. Indicators included: repeated complaints from female engineers; disparate promotion/hiring statistics; exit interview data showing gendered experiences; missed defect detections linked to lack of diverse perspectives; and deviations between stated non-discrimination policies and actual outcomes. These signals raised moral salience because they pointed to predictable harms (career harm to employees, possible safety risks for end users) and violated norms of fairness. Recognizing disparate impact and connection to product safety is the first step toward ethical action. [asme.org+1](#)

Moral Reasoning:

Analysis of choices using ethical reasoning.

- ❖ From a legal/organizational viewpoint, bias may expose the firm to employment-law claims and violate internal policies; legal compliance alone is insufficient as a moral guide.
- ❖ From a safety-first perspective, engineers' primary obligation is to the safety, health, and welfare of the public; systemic exclusion that degrades product quality undermines that paramount duty.
nspe.org

Right vs. beneficial vs. legal.

- ❖ *Right*: Acting to correct unfair treatment and ensure equitable opportunities is morally required by principles of fairness and respect for persons.
- ❖ *Beneficial*: Structural pro-diversity measures are likely to produce better technical outcomes (innovation, defect detection) and economic benefits (organizational performance). [McKinsey & Company](#)
- ❖ *Legal*: Some structural steps could raise concerns if framed as quotas, so implementation must be legally reviewed and based on transparent, merit-based criteria that remediate bias.

Justification (clear logic). The firm's obligation to public safety (NSPE/IEEE canons) and to treat persons fairly (IEEE/NSPE/ASME guidance) create a duty to remedy discriminatory practices that impair product quality and employee dignity. Because harm to patients and employees is foreseeable if exclusion continues, the precautionary and duty-based arguments require corrective action beyond tokenism.
dusk.geo.orst.edu+1

Ethical Theories Applied:

Utilitarianism (consequentialist).

A utilitarian analysis asks which option maximizes overall well-being (patients, staff, shareholders, society). Implementing structural diversity measures (Option B) likely increases team performance, reduces product failures, retains talent, and improves societal trust—producing the greatest net good. Empirical business research links diversity in leadership to better organizational outcomes, supporting this consequence-based choice. [McKinsey & Company+1](#)

Deontology (duty-based).

Deontological ethics emphasizes obligations: engineers must hold public safety paramount and treat people with fairness and respect. Codes require honesty, non-discrimination, and duty to report unethical practice. These duties mandate that NovaTech act to remove bias and protect employees and the public, independent of consequences. [dusk.geo.orst.edu+1](#)

Virtue Ethics (character and integrity).

Virtue ethics focuses on moral character—integrity, justice, courage. Leadership that demonstrates courage to confront bias and integrity to change institutional norms embodies virtuous engineering practice. Cultivating an organizational culture of inclusion is an exercise in institutional virtue. [asme.org](#)

Rights-Based Ethics.

Engineers and applicants have rights to equal treatment and nondiscrimination. Patients have rights to safe medical devices. Rights-based reasoning supports remedies that protect individual and user rights (e.g., fair hiring, inclusive testing). [nspe.org](#)

Justice Ethics (fairness).

Justice demands fair distribution of opportunities and burdens. When some groups bear the burden of exclusion, justice requires correcting imbalances (equity, not just equality). This supports targeted measures that remove barriers and ensure fair access to advancement. [asme.org](#)

How theory guides decision-making.

Taken together, these theories converge on proactive, structural remediation (Option B): it respects duties and rights, promotes virtues, and produces better outcomes.

Ethical Codes and Standards :

IEEE Code of Ethics. The IEEE Code requires members to accept responsibility for decisions consistent with public safety, to improve understanding of technology, and to “treat fairly all persons regardless of ... gender.” These principles support corrective action to remove gender bias that could endanger the public or employees. dusk.geo.orst.edu

NSPE Code of Ethics for Engineers. The NSPE Code’s Fundamental Canons instruct engineers to hold the public welfare paramount, act with fairness and equity, and conduct themselves honorably. Engineers aware of biased practices that threaten product quality or employee welfare have an obligation to report and correct them. nspe.org

ASME diversity and inclusion policies. ASME’s policy documents and society values explicitly endorse diversity, equitable access, and inclusive decision-making, offering organizational best practices that engineering employers can model. asme.org

UN Sustainable Development Goal 5 (Gender Equality). Broader social commitments such as SDG 5 (achieve gender equality and empower all women and girls) provide an international ethical framework reinforcing organizational action on gender issues. SDGs

Implication. Professional codes and global goals make clear that both individual engineers and organizations share obligations to remediate discriminatory practices that risk harm to people or degrade professional integrity.

Conclusion and Recommendations:

Findings and moral position. The pattern of biased hiring and promotion at NovaTech creates foreseeable harms to employees (career damage, psychological harm) and to the public (reduced product robustness for a safety-critical medical device). Ethical analysis—drawing on utilitarian, deontological, virtue, rights, and justice perspectives—points to the obligation to implement robust, structural remedies rather than superficial measures.

Best ethical decision (recommended). NovaTech should adopt Option B (structural interventions) implemented with legal safeguards and transparent metrics. Key elements:

1. Immediate bias audit and disclosure. Commission an internal/external audit of hiring, promotion, assignment, pay, and performance data; publish results transparently to employees and (internally) to stakeholders. This follows professional obligations to acknowledge and correct harmful practices. [nspe.org](https://www.nspe.org)
2. Revise hiring & promotion processes. Implement anonymized CV screening where feasible, standardized evaluation rubrics, diverse interview panels, explicit job criteria, and documented rationale for promotion decisions. Link promotions to objective performance and leadership competencies. [asme.org](https://www.asme.org)
3. Accountability metrics and targets (not rigid quotas). Set time-bound representation and inclusion targets tied to leadership performance metrics; report progress publicly at agreed intervals. Targets should be legally reviewed and framed as equitable corrective measures. [McKinsey & Company](https://www.mckinsey.com)
4. Inclusive design and testing requirements for safety-critical systems. Mandate diverse representation on design-review and human-factors teams; require test plans that explicitly account for diverse user needs and failure modes. This protects public welfare as required by NSPE/IEEE canons. [nspe.org+1](https://www.nspe.org+1)

5. Mentoring, sponsorship, and career development with protected time. Provide structured sponsorship for underrepresented engineers and ensure equitable access to high-visibility assignments.
6. Training plus enforcement. Combine unconscious-bias and inclusive-leadership training with enforceable policies (anti-retaliation, clear complaint processes, independent ombuds). Training alone is insufficient—pair it with consequences for discriminatory conduct. asme.org
7. Cultural change program led from the top. Senior leaders must model inclusive behavior, commit resources, and integrate DEI goals into business strategy.

How this avoids similar problems. These steps remove structural barriers, improve detection of design shortcomings (improving safety and product quality), retain talent, and align the firm with professional ethical codes and societal commitments (SDGs).

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